



**VERIFYNOW
SYSTEM**

Jubail Industrial College

Computer Science & Information Technology Department (CSIT)

Final Report

Project Title: VerifyNow App

Group #: 7

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1. INTRODUCTION

The rapid growth of social media platforms has made it easier for users to share information instantly. However, this has also led to the widespread dissemination of misinformation. False information often spreads faster than verified facts, while many users lack the time, effort, or accessible tools needed to verify the accuracy of the content they consume or share. As a result, misinformation continues to circulate unintentionally, negatively impacting public awareness and decision-making.

1.1 Project Proposal Summary

To address this problem, the VerifyNow system is proposed as a solution that enables users to verify information before sharing it. The system is designed to simulate human-like verification by analyzing content based on source credibility, context, and supporting evidence.

The system consists of several main components, including an input module that allows users to submit content, a verification engine that evaluates the information against reliable sources, and a source and context analysis module that examines how and why the information was presented. In addition, a user-friendly interface is provided to display results clearly, along with a learning and enhancement feature that improves the system over time based on user feedback.

The primary users of the system are general social media users who want to ensure the accuracy of the information they share. The system can also be beneficial for journalists, researchers, and organizations that require quick and reliable verification tools. Compared to existing solutions, which often require users to manually search for information or provide general results, VerifyNow offers a dedicated verification process that combines automated analysis with human-like reasoning. This enables users to obtain faster, clearer, and more reliable verification results within a single platform.

1.2 Risk Management

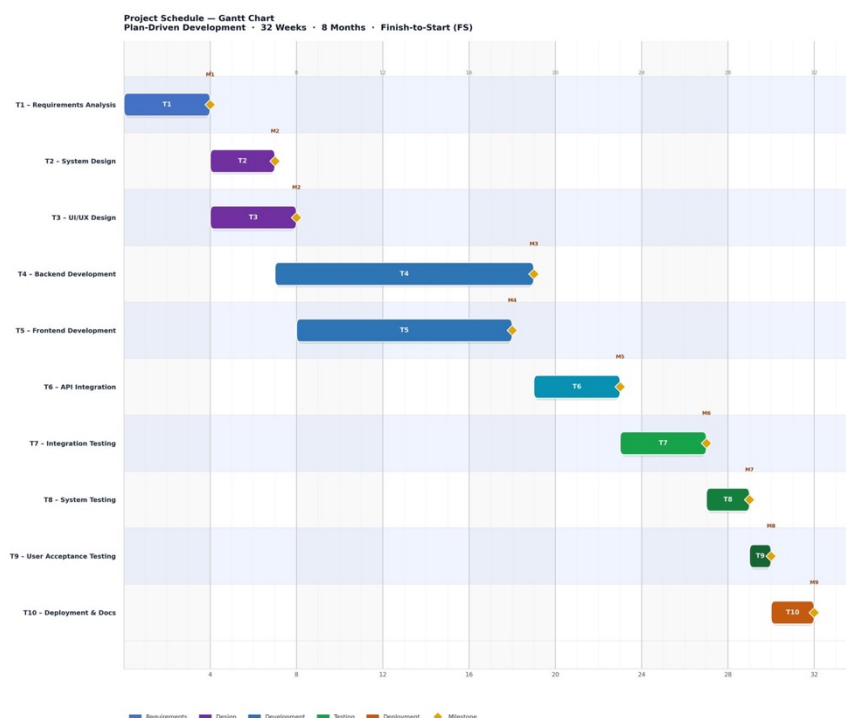
The project has been carefully planned to ensure reliability and successful completion. Several potential risks have been identified, including inaccurate verification results, dependency on external data sources, legal concerns related to content classification, and possible system downtime. Appropriate mitigation strategies have been defined to minimize these risks and maintain system performance. In addition, system monitoring, backup solutions, and the use of multiple verification sources will help improve the reliability and availability of the application.

Risk ID	Risk Name	Type	Description	Affected	Probability	Consequences	Strategy	Remediation Plan	Alternative Plan	Monitoring
R1	Inaccurate Verification Result	Technical	The system may produce an incorrect result due to limitations in artificial intelligence. For example, a true but outdated piece of news might be classified as a rumor by the system because it relies only on recent data.	Product	Moderate	Serious	Minimize	Improve analysis models and use multiple sources. Display the confidence level and warn of uncertain results.	Display a message indicating that the investigation is currently unavailable.	- Periodically measure the accuracy rate of verification results. - Compare system results with reliable verification sources. - Review complaints or objections related to inaccuracies in the results.
R2	Reliance on External APIs	External	Verification may stop if external validation interfaces are unavailable or their policies change. For example, Google Fact Check and Tavily APIs may stop or fail.	Product	Moderate	Tolerable	Minimize	Use and update multiple APIs.	Display a message indicating that the investigation is currently unavailable.	- Continuously monitor the availability and response of the APIs. - Track the number of connection failures or service interruptions. - Regularly review updates or changes in the policies of external service providers.
R3	Legal and Defamation Risk	Legal & Ethical	The application may classify a news article, organization, or individual statement as false or misleading, which could result in legal complaints or defamation claims. If the system incorrectly labels content as fake, affected parties may take legal action against the project.	Business & Product	Moderate	Catastrophic	Minimize	Add clear disclaimer stating that results are based on automated analysis. Avoid definitive labels such as "Fake" and instead use "Likely inaccurate" or "Unverified. Consult legal guidelines related to digital publishing. Store verification logs for transparency.	If a complaint is received, temporarily remove the flagged content and conduct manual review before re-publishing the result.	- Track legal complaints or warning notices. - Periodically review system decisions for sensitive topics. - Monitor user reports related to unfair classification.
R4	Inconsistent Human-Like Verification	Technical	The system is designed to simulate how humans evaluate information by analyzing context, source credibility, tone, and supporting evidence. However, translating human reasoning into programmed rules is challenging. If the evaluation criteria are incomplete or poorly calibrated, the system may produce inconsistent results when analyzing similar content, leading to reduced reliability and user trust.	Product	Moderate	Serious	Minimize	Define clear verification criteria and validation guidelines. Test the system using real case studies before deployment.	Allow disputed results to be flagged for manual expert review and adjust the verification logic accordingly.	- Measure consistency rate between similar verification cases. - Track number of user disputes or corrections.
R5	System Downtime and Unavailability	Operational	There is a risk that the application will have downtime or unavailability.	Business & Product & Project	High	Serious	Minimize	Redundancy will be implemented to ensure that the application will have high availability. The performance of the servers will be tested and monitored. A contract will be developed for the performance of the servers.	If there is downtime or unavailability of the application, alternative methods will be provided for the users. The users will also be notified.	- The efficiency and availability of the server will be monitored in real time. - The results ports from downtime will be tracked. - The server will be possible vulnerability.
R6	Integration Complexity Between Modules	Project	VerifyNow includes multiple modules (input, Verification Engine, source analysis, and output). Integration between these components may cause delays or functional conflicts.	Project	Moderate	Tolerable	Minimize	Adopt modular development with clear APIs between modules. Perform integration testing early, not only at the final stage.	Temporarily disable non-critical modules to stabilize the core system.	- Number of integration defects discovered during testing. - Time required to connect modules successfully.
R7	Limited Resources and Time Constraints	Organizational	Since the project is developed by students using standard hardware and limited budget, there may be constraints in time, tools, or technical expertise.	Project & Business	High	Tolerable	Accept	Focus on delivering a Minimum Viable Product (MVP) that includes only essential verification features. Use open-source tools to reduce cost and effort. Divide responsibilities clearly among team members to improve efficiency.	Adjust schedule, reduce secondary features and extend development into future phases after the specific submission.	- Compare planned versus actual task completion rates. - Weekly progress tracking and milestone reviews.

1.3 Project Schedule

The project follows a plan-driven development approach with an estimated duration of eight months. The development process is divided into four main phases: requirements analysis, design and development, testing, and deployment. Requirements analysis represents 15% of the total project duration, while design and development represent 55%. Testing activities account for 20% of the schedule, followed by deployment and documentation with 10%. This structured schedule ensures that the system is developed in an organized manner and meets the required quality standards.

Task ID	Task Name	Short Description	Duration	Assigned Members
T1	Requirements Analysis	Define functional & non-functional requirements and prepare SRS document	4 Weeks	2 Members
T2	System Design	Design system architecture, modules, and database structure	3 Weeks	2 Members
T3	UI/UX Design	Design application screens and user experience	4 Weeks	2 Members
T4	Backend Development	Develop verification engine and core system logic	12 Weeks	3 Members
T5	Frontend Development	Develop mobile app interface and connect to backend	10 Weeks	3 Members
T6	API Integration	Integrate fact-checking and AI APIs into the system	6 Weeks	2 Members
T7	Integration Testing	Test interaction between frontend, backend, and APIs	4 Weeks	2 Members
T8	System Testing	Perform full system testing and fix issues	4 Weeks	3 Members
T9	User Acceptance Testing	Collect feedback from users before release	2 Weeks	2 Members
T10	Deployment & Documentation	Deploy system and prepare final documentation	2 Weeks	2 Members



1.4 Proposed Budget

An estimated budget has been allocated for the development of the Minimum Viable Product (MVP) over a period of 6–9 months. The budget is divided into several main components, including app development (Android/iOS) using technologies such as Flutter estimated at \$6,000–\$12,000, verification system development (API and source analysis) estimated at \$5,000–\$7,000, hosting and cloud services estimated at \$1,500–\$2,500, UI/UX design estimated at \$2,000–\$2,500, and testing, optimization, and initial maintenance estimated at \$1,000–\$1,500. The approximate total project budget ranges from \$15,500 to \$25,000.

Overall, VerifyNow aims to reduce the spread of misinformation by providing an accessible, efficient, and reliable verification system that supports users in making informed decisions in the digital environment.

2. REQUIREMENTS SPECIFICATION (SRS)

This section presents the Software Requirements Specification (SRS) for the VerifyNow system. It outlines the Information Seeker and system requirements needed to verify information from online links.

User Requirement 1:

1. Information Seekers should be able to enter a link so they can verify the information it contains.

System Requirements:

Functional Requirements

1.1 Link Input

The VerifyNow system shall let users enter a URL to verify the information it contains.

1.2 Verification

The VerifyNow system shall verify the information in the link.

1.3 Display Verification Result

The VerifyNow system shall show Information Seekers the result of the verification.

User Requirement 2:

2.Information Seekers should be able to see the confidence level of the verification result so they can understand how reliable the result is.

System Requirements

Functional Requirements

2.1 The VerifyNow system shall calculate a confidence level for the verification result.

2.2 The VerifyNow system shall display the confidence level to Information Seekers.

2.3 The VerifyNow system shall display the confidence level as a percentage

(for example: 85%).

User Requirement 3:

3.Information Seekers should be able to receive a warning if the information in the link is misleading or untrustworthy in order to avoid spreading false information.

System Requirements

Functional Requirements

3.1 The VerifyNow system shall identify whether the information is misleading or untrustworthy.

3.2 The VerifyNow system shall display a warning message if the information is identified as misleading.

Non-Functional Requirements

3.3 Reliability

The VerifyNow system shall provide accurate warnings based on trusted sources of information.

User Requirement 4:

4.Information Seekers should be able to use the system easily without requiring technical knowledge so that anyone can verify information quickly.

System Requirements

Functional Requirements

4.1 The VerifyNow system shall provide a simple interface for Information Seekers to enter and verify links.

4.2 The VerifyNow system shall allow Information Seekers to submit a link with minimal steps (e.g., clicking a verify button).

User Requirement 5:

5.Information Seekers should be able to report incorrect verification results so the system can improve its accuracy.

System Requirements

Functional Requirements

5.1 The VerifyNow system shall provide a “Report Result” option.

5.2 The VerifyNow system shall allow Information Seekers to submit feedback on incorrect results.

5.3 The VerifyNow system shall store the reported feedback for system review and improvement.

User Requirement 6:

6.Information Seekers should be able to see a short explanation of why the system classified the information as true or misleading so they can better understand the verification result.

System Requirements

Functional Requirements

6.1 The VerifyNow system shall generate a brief explanation describing how the verification result was determined.

6.2 The VerifyNow system shall highlight key statements used in the verification process.

6.3 The VerifyNow system shall display the explanation along with the verification result.

3. SOFTWARE DESIGN

This section presents the software design of the VerifyNow system. It includes different system models such as the context model, use case diagram, sequence diagrams, and class diagram. These models describe both the structure and behavior of the system and provide a clear understanding of how VerifyNow operates.

3.1 Context Model

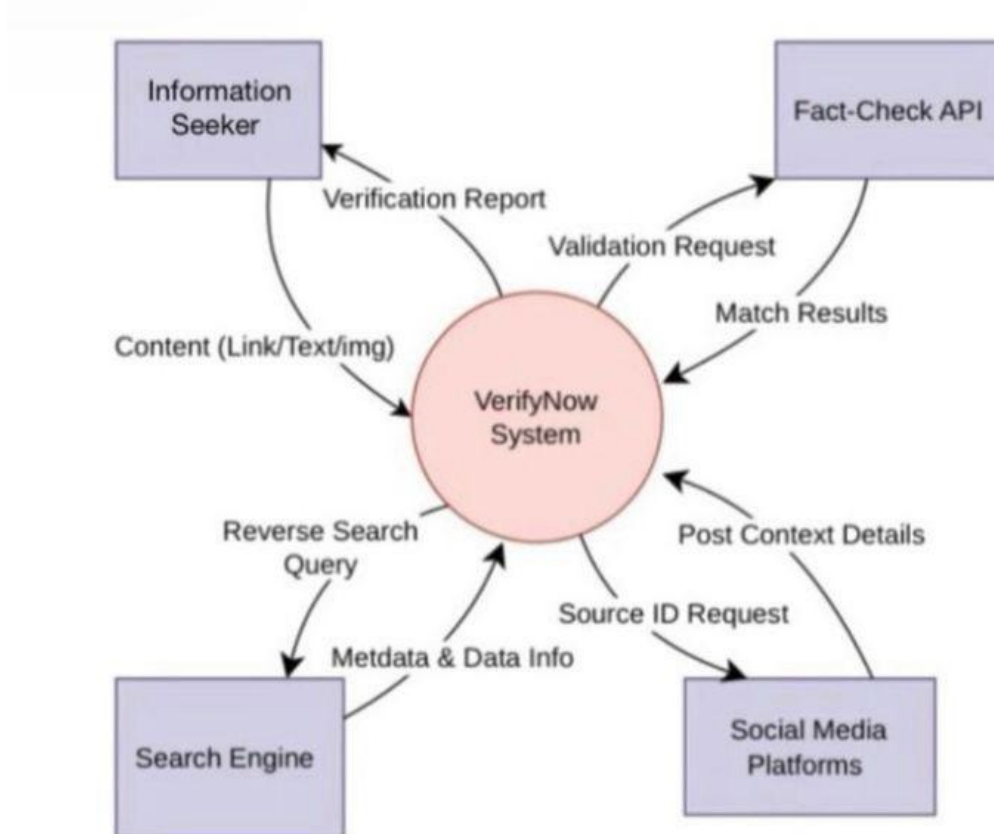


Figure 3.1: VerifyNow Context Model

The Context Model illustrates the overall environment of the VerifyNow system and defines its boundaries. It shows how the system interacts with external entities such as the Information Seeker, Fact-Check APIs, Search Engines, and Social Media Platforms. The Information Seeker submits content (links, text, or images) to the system for verification, and the system returns a verification report indicating the accuracy of the information. External services play a key role in supporting the system by providing fact-checking data, metadata, and contextual information. This model helps in understanding the high-level data flow between the system and its surrounding environment.

3.2 Use Case Diagram

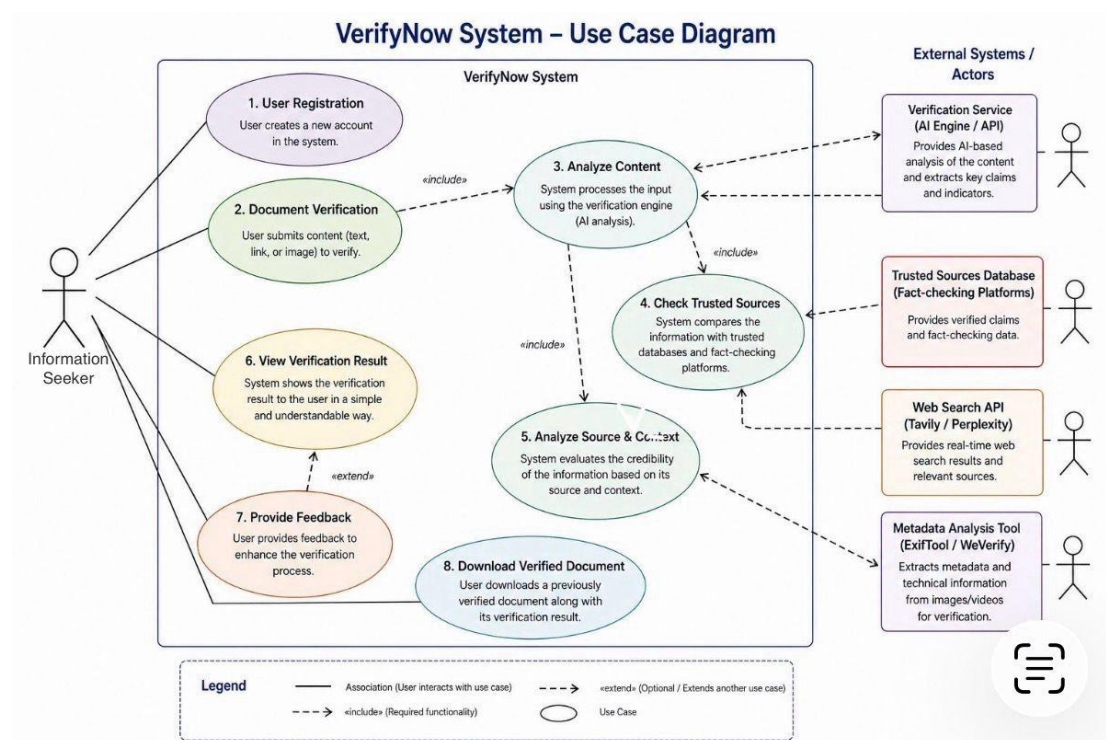


Figure 3.2: VerifyNow Use Case Diagram

The Use Case Diagram represents the functional requirements of the VerifyNow system by illustrating the interaction between the primary actor (Information Seeker) and the system functionalities. The main use cases include User Registration, Document Verification, Viewing Verification Results, Providing Feedback, and Downloading Verified Documents. After the Information Seeker performs the Document Verification process, the system executes internal processes such as analyzing content, checking trusted sources, and evaluating context to ensure accurate results. The diagram also shows the interaction between the system and external services such as verification APIs and data sources. Overall, it provides a clear overview of how the Information Seeker interacts with the system and how the system fulfills the required functionalities.

3.3 Detailed use case and sequence diagram:

Use Case 1: User Registration

Actor: information seeker

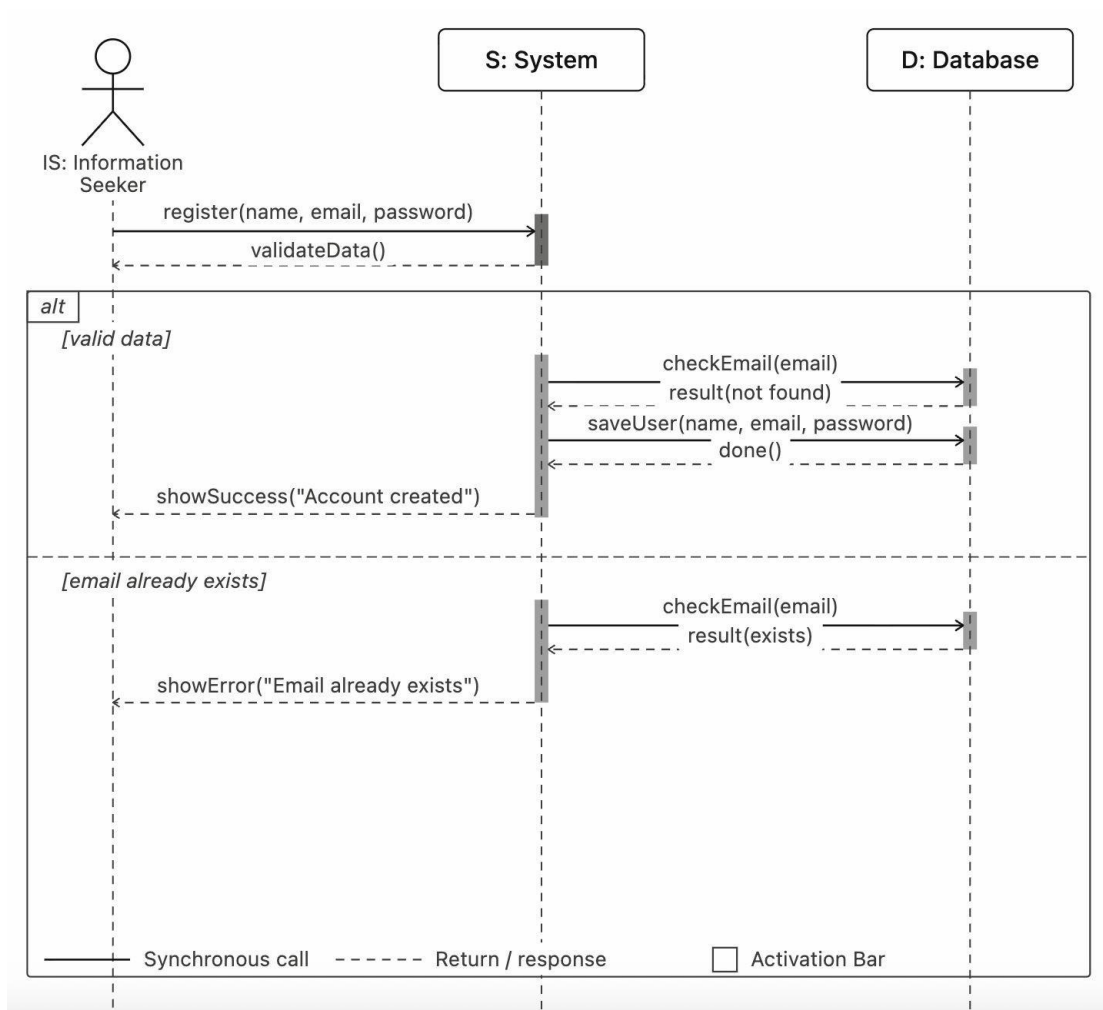


Figure 3.3: VerifyNow User Registration Sequence Diagram

This Sequence Diagram illustrates the process of user registration in the VerifyNow system. It shows the interaction between the Information Seeker, the System, and the Database. The process begins when the Information Seeker submits registration information through the interface. The system then validates the input data and checks whether the email already exists. If the data is valid, the system stores the new user information in the database and confirms successful account creation. This sequence ensures a secure and accurate registration process.

Use Case 2: Document Verification

Actor: information seeker

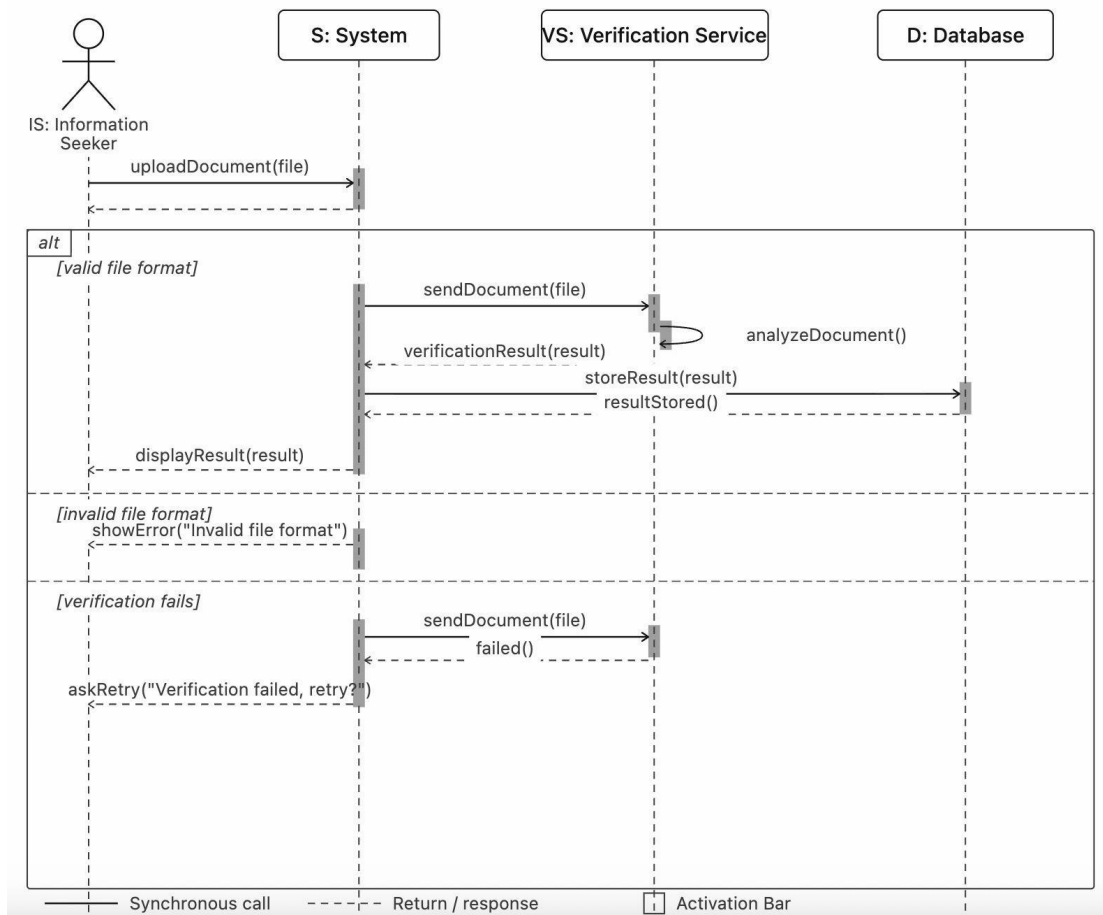
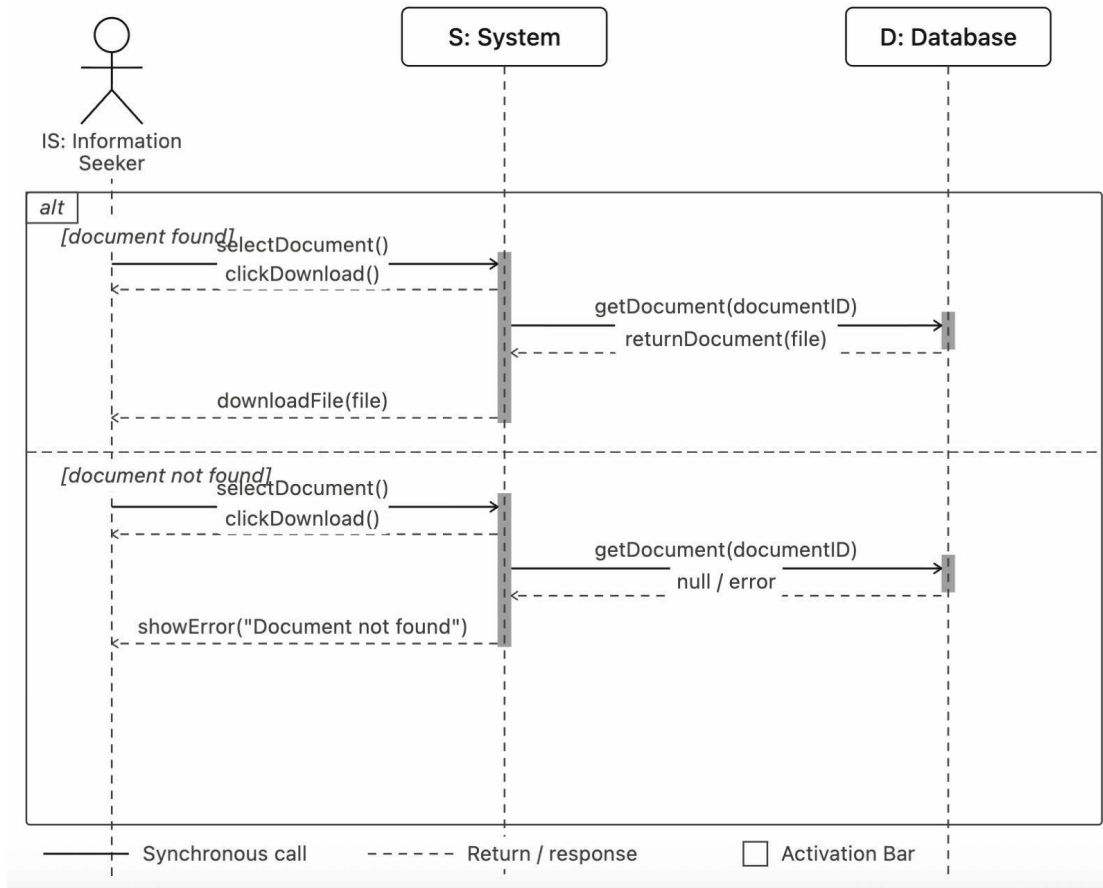


Figure 3.4: VerifyNow Document Verification Sequence Diagram

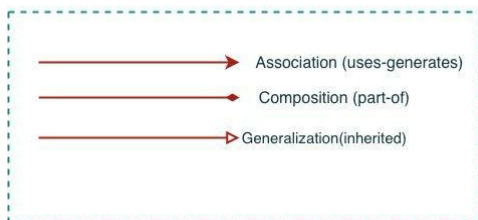
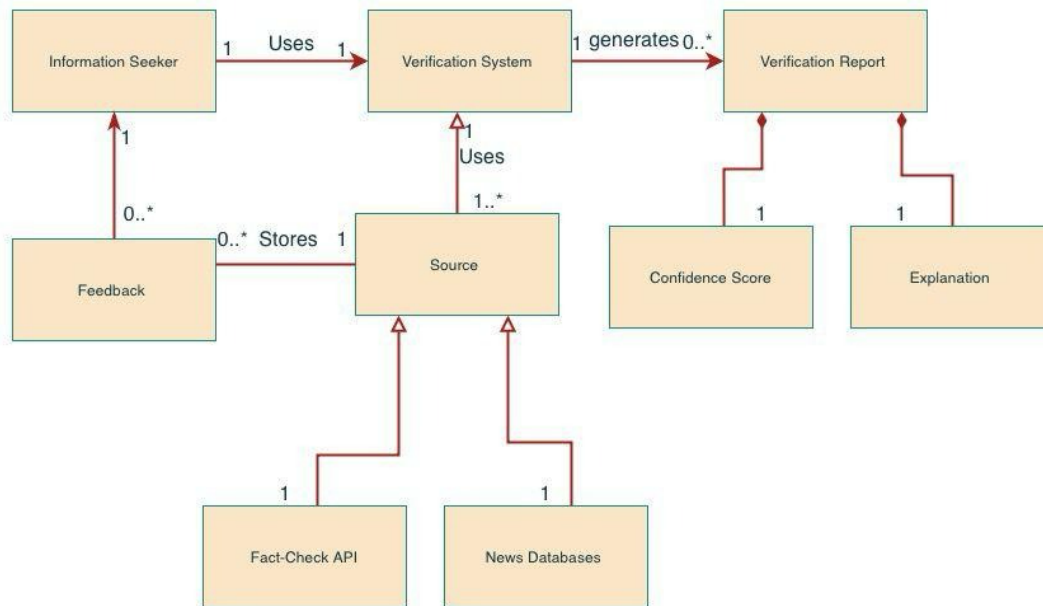
This Sequence Diagram represents the document verification process in the VerifyNow system. It demonstrates how the Information Seeker interacts with the System and external verification services. The process starts when the Information Seeker uploads a document or submits content for verification. The system processes the input and sends it to external verification services such as fact-checking APIs and search engines. After analyzing the data, the system receives the verification result, stores it, and displays it to the Information Seeker. This process ensures reliable and accurate verification of content.

Use Case 3: Download Verified Document**Actor:** information seeker**Figure 3.5: VerifyNow Download Verified Document Sequence Diagram**

This Sequence Diagram illustrates the process of downloading a verified document. It shows the interaction between the Information Seeker, the System, and the Database. After a document has been verified, the Information Seeker requests to download it. The system first checks the verification status to ensure the document is valid. Once confirmed, the system retrieves the document from the database and sends it to the Information Seeker. This ensures that only verified and trusted documents are accessible for download.

3.4 High Level Class Diagram

High level class diagram for VerifyNow System



Explanation:

- The Information Seeker uses the Verification System to verify a link.
- The Verification System uses different Sources to analyze the link and generates one or more Verification Reports.
- Each Verification Report is composed of a Confidence Score and an Explanation.
- Source is a general class with two types: Fact-Check API and News Database.
- Information Seekers can submit Feedback, which is stored by the system.

Figure 3.6: VerifyNow high level Class Diagram

The High-Level Class Diagram provides an overview of the main classes in the VerifyNow system and their relationships. It identifies the key components such as Information Seeker, Document, Verification, and System, and shows how these classes interact with each other at an abstract level.

This diagram focuses on the overall structure of the system without going into detailed attributes and methods, helping to understand the system architecture and main responsibilities of each class.

3.5 Detailed class diagram

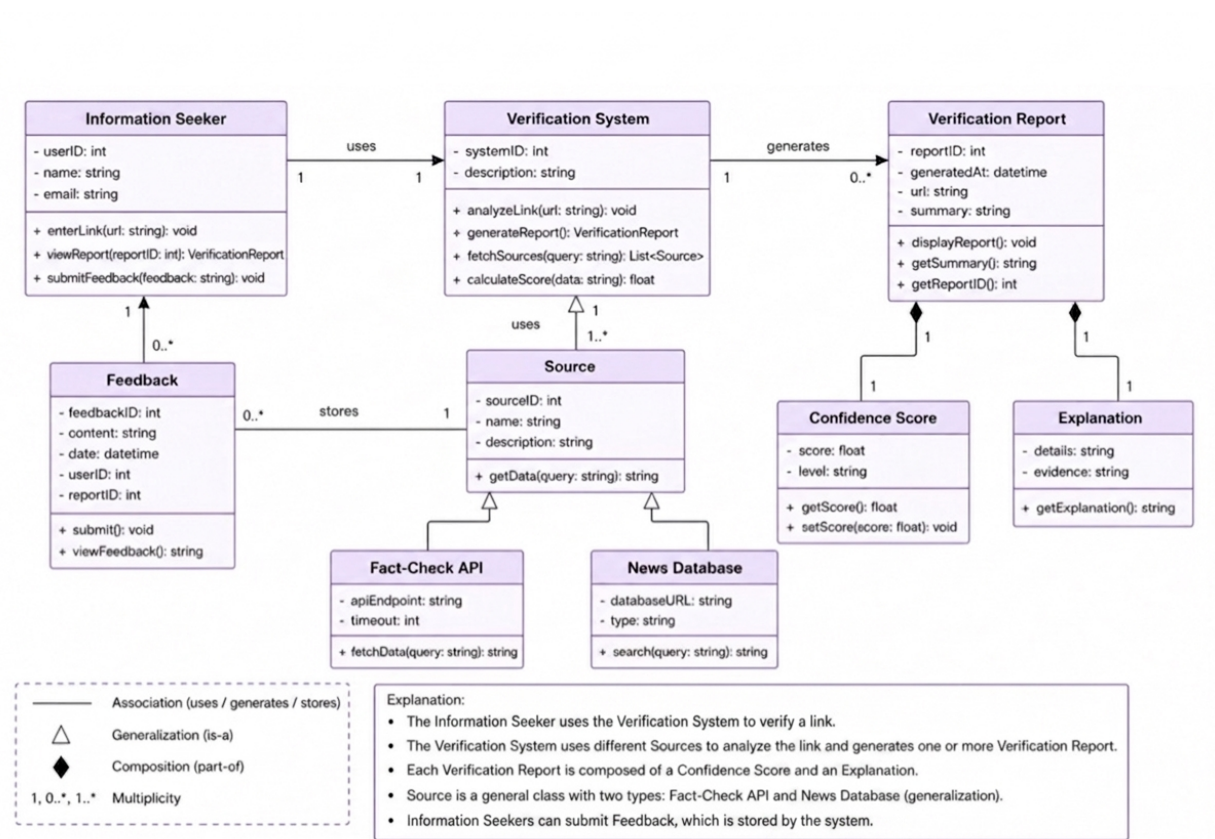


Figure 3.7: VerifyNow Detailed Class Diagram

The Detailed Class Diagram presents the internal structure of the VerifyNow system in detail. It includes the main classes such as Information Seeker, Document, Verification, and System, along with their attributes and methods. The diagram also illustrates the relationships between classes, including associations and dependencies. For example, the Information Seeker interacts with the Document class to upload and retrieve documents, while the Verification class is responsible for processing and validating content. This diagram serves as a blueprint for system implementation and provides a clear understanding of how different components work together.

4. SOFTWARE IMPLEMENTATION

4.1 Development Model

This project follows a Plan-Driven development approach. The Plan-Driven model was selected because the project focuses heavily on system analysis, documentation, requirements engineering, and software design before implementation. This approach ensures that all project phases are carefully planned and documented in an organized manner.

4.2 Development Process

The project team worked remotely and conducted regular online meetings to discuss project progress, assign tasks, and review system requirements and design models. Since there were no external clients involved in the project, user requirements and system needs were identified and analyzed internally by the team.

In addition, the SRS document and software design models were reviewed collectively to ensure consistency, correctness, and alignment with the project objectives.

4.3 Tools and Environment

The VerifyNow project is designed as a software-based conceptual web system and does not require specialized hardware components. Standard personal computers and laptops were sufficient for development, documentation, and testing activities.

The following tools and technologies were used during the project development process:

- Hardware: Personal computers and laptops
- Platform: Web-based conceptual system
- Development Tools: Visual Studio Code
- Communication Tools: Microsoft Teams and Zoom
- Network Environment: Internet connection for communication and external API access
- APIs: Fact-checking APIs to support content verification

These tools were selected because they are reliable, accessible, and suitable for collaborative software development and communication.

4.4 Testing and Validation

Since the VerifyNow project mainly focuses on system analysis and software design, testing and validation activities were conducted through design reviews, requirement analysis, and evaluation of potential system issues rather than full implementation testing.

During the review process, several possible issues were identified, including unclear user input, inaccurate classification of information, and dependency on external verification sources. To reduce these risks, several validation approaches were proposed. These approaches include input validation to ensure that users provide clear and valid information, improving the verification logic to increase classification accuracy, and integrating external fact-checking APIs to support reliable verification results.

In addition, the SRS document and software design models were reviewed collectively by the team to ensure consistency, correctness, and alignment with the project objectives. These validation activities helped improve the overall reliability and quality of the proposed system design.

4.5 Test Cases

Even though the system has not been fully implemented, the following test cases were designed to demonstrate how the system functionality could be validated in future implementations.

Test Case 1:

- Input: User submits a reliable news link
- Expected Result: The system returns “Verified” and displays the verification result successfully.

Test Case 2:

- Input: User submits false or misleading information
- Expected Result: The system returns “Not Verified” and displays a warning message to the user.

5. Conclusion

This project focused on analyzing and designing the VerifyNow system as a solution to help users verify online information before sharing it. During the project, several software engineering activities were completed, including requirements analysis, system modeling, risk management, and project scheduling.

5.1 System Status

At the current stage, the system design and documentation were completed successfully, including the SRS document, UML diagrams, and project planning activities. However, the project has not yet reached the full implementation stage, and some features still require further development and integration with external services.

5.2 Future Enhancement

Future improvements may include developing a complete mobile application, improving verification accuracy, supporting additional verification sources, and enhancing the user interface and user experience.

5.3 Professional Development and Lessons Learned

This project helped the team gain practical experience in requirements engineering, UML modeling, teamwork, task distribution, and software documentation. It also improved communication and coordination skills between team members throughout different project phases.

5.4 Challenges and Obstacles

One of the main challenges was maintaining consistency between different system models and diagrams. The team also faced difficulties related to selecting suitable APIs and organizing project tasks within the available time. These challenges were managed through continuous reviews, discussions, and regular updates to the project documents.

Overall, the project provided valuable experience in applying software engineering concepts to a real-world problem.